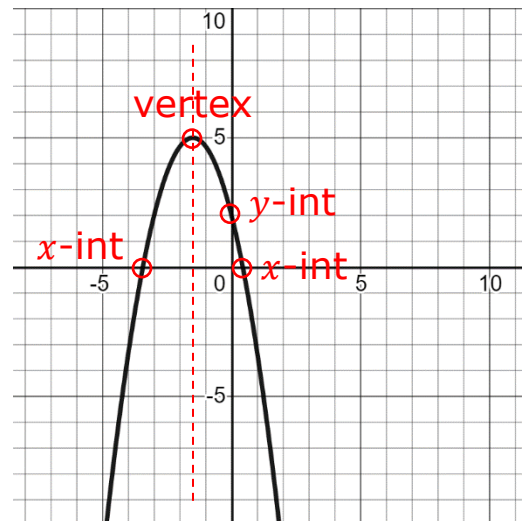


Algebra 1
Unit 8
Lesson 1 Homework

Name **Answer Key** _____
Date _____
Period _____

Use the graph of the quadratic equation for questions 1 – 4.

- Describe where the parabola is increasing.
 $(-\infty, -1.5), x < -1.5$
- Describe where the parabola is decreasing.
 $(-1.5, \infty), x > -1.5$
- Draw the axis of symmetry for the parabola on the graph.



- Circle the vertex, x -, and y - intercepts of the parabola. Label each point as vertex, x -intercept, and y - intercept.
- The equations $y = 4(x - 1)(x - 1)$ and $y = 4(x - 1)^2$ both represent the same quadratic. Rewrite both equations in standard form, $y = ax^2 + bx + c$. Show your work.

$y = 4(x - 1)(x - 1)$	$y = 4(x - 1)^2$
$y = (4x - 4)(x - 1)$ $y = 4x^2 - 4x - 4x + 4$ $y = 4x^2 - 8x + 4$	$y = 4(x^2 - 2x + 1)$ $y = 4x^2 - 8x + 4$

- The equations $y = -(x + 7)(x + 4)$ and $y = -(x - (-7))(x - (-3))$ are equivalent. How can both x -intercepts be determined if the equation $y = -(x + 7)(x + 4)$ is given?

$$x + 7 = 0$$

$$x = -7$$

$$x + 4 = 0$$

$$x = -4$$

Set each factor equal to 0 and solve.

7. The equation $y = \frac{1}{3}x^2 + x - 18$ can be used to determine the y -intercept. Show how each of the *three* forms can be used to find the y -intercept.

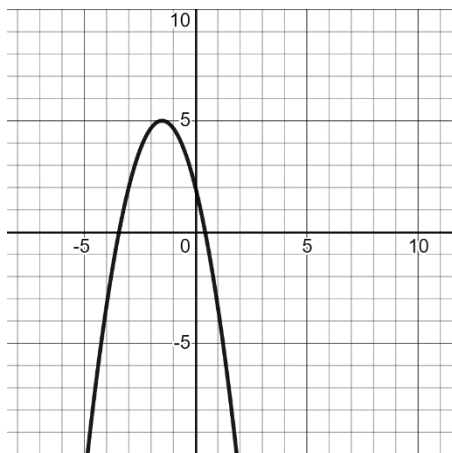
$y = \frac{1}{3}x^2 + x - 18$	$y = \frac{1}{3}(x + 9)(x - 6)$	$y = \frac{1}{3}\left(x + \frac{3}{2}\right)^2 - \frac{225}{12}$
$y = \frac{1}{3}(0)^2 + 0 - 18$ $y = -18$	$y = \frac{1}{3}(0 + 9)(0 - 6)$ $y = \frac{1}{3}(9)(-6)$ $y = -18$	$y = \frac{1}{3}\left(0 + \frac{3}{2}\right)^2 - \frac{225}{12}$ $y = \frac{3}{4} - \frac{225}{12}$ $y = -18$

8. A table of values for $y = -\frac{2}{3}x^2 + \frac{32}{3}x - 42$ is shown. What features can be found using the table of values?

- ☒ x -intercepts
☐ y -intercepts
☒ vertex
☒ axis of symmetry
☒ increasing/decreasing
☒ positive/negative

x	y
4	-10
5	-5.333
6	-2
7	0
8	0.667
9	0
10	-2

9. The graph of a quadratic equation is shown. Use words, ordered pairs, or symbols to describe the location of each key feature.



Key Feature	Location of Key Feature
x -intercepts	$(-3.5, 0)$ $(0.5, 0)$
y -intercepts	$(0, 2)$
vertex	$(-1.5, 0.5)$
axis of symmetry	$x = -1.5$
increasing	$(-\infty, -1.5), x < -1.5$
decreasing	$(-1.5, \infty), x > -1.5$

10. A table of values for a quadratic equation shown. What features can be found using the table of values?

- ☐ x -intercepts
☒ y -intercepts
☐ vertex
☐ axis of symmetry
☐ increasing/decreasing
☐ positive/negative

x	y
-1	16
0	15
1	12
2	7